

Occurrence of antibiotic-resistant bacteria in urban surface water sources in Bangladesh

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Abstract

Infections caused by antibiotic-resistant bacteria (ARB) result in thousands of human deaths annually worldwide. Surface waters are impacted by anthropogenic factors, which contribute to the emergence and spread of ARB in the aquatic environment. There has been a notable lack of study on antibiotic resistance in surface water, particularly in developing nations like Bangladesh, where antibiotics are widely consumed and are not disposed properly. In this study, bacteria strains isolated from three rivers and two lakes in Khulna city, Bangladesh were characterized for their antibiotic resistance using disc diffusion method. Overall, of 56 isolates of bacteria from samples of the surface water sources, most were resistant to Ciprofloxacin (75.0–87.5%) and Ceftriaxone (65.6–78.1%). Ampicillin showed (9.4–18.8%) a comparatively lower resistance rate than that of Ciprofloxacin and Ceftriaxone. The prevalence of ARB was observed to be higher during the wet seasons compared to the dry seasons. The 16S rRNA gene analysis showed that *Shigella flexneri* was dominant (17.9%) in surface water followed by *Escherichia fergusonii* (12.5%), *Proteus mirabilis* (10.7%) and *Enterobacter quasihormaechei* (8.9%). The genus level analysis showed that *Enterobacter* (23.5%), *Shigella* (20.6%), and *Escherichia* spp (14.7%) were found most abundant both in the rivers and lakes water. The findings of this study underscore the urgent need for routine surveillance of antimicrobial resistance in surface water sources nationwide, as well as the implementation of safe disposal practices for antibiotics utilized in healthcare, animal husbandry, and aquaculture.

1. Introduction

Antibiotics are highly effective in treating bacterial diseases. However, increasing wastewater discharge and the emergence of antibiotic-resistant bacteria (ARB) in aquatic environment pose significant concerns for human health in recent years (Anderson et al., 2013; Yuan et al., 2015; Islam 2019). Aquatic environment (e.g. lakes, rivers, canals) receives effluent from municipalities and industries, as well as runoff from urban and agricultural areas. These inputs are substantially linked to the elevated natural levels of ARB (Islam, 2019; Mahmood et al., 2019). Antibiotic-resistant genes (ARGs) have been identified in aquatic environments due to widespread use of antibiotics in human and veterinary medicine

(Neela et al., 2007; Islam, 2019), and aquatic environment is likely the ultimate reservoir for drug resistant bacteria (Neela et al., 2015). Occurrences of many ARB and ARGs have been frequently reported in sewage, treated drinking water and surface water (Yuan et al., 2015). If resistant bacteria are found in surface water (e.g. river, streams, lake, pond) that people use frequently for bathing, recreation or drinking water production can cause substantial health risk and disease burden to the people. Infections caused by ARB result in thousands of human deaths every year worldwide (Kusi et al., 2022). ARB poses the capability to transmit resistance to antibiotics and reduce the efficacy of antibiotics within the aquatic systems where they proliferate (Islam, 2019). Antibiotic resistance may occur in nature, but misuse/overuse and improper disposal of antibiotics are accelerating this process (Islam, 2019; WHO, 2020).

Conventional Wastewater Treatment Plants (WWTPs) cannot fully remove ARB and ARGs from the final treated effluent (Hong et al., 2018). Chlorination is not effective in controlling antimicrobial resistance. Yuan et al. (2015) found that about 40% of erythromycin-resistance genes and 80% of tetracycline resistance genes could not be removed by chlorination. ARGs can be transferred between organisms in aquatic environments through horizontal gene transfer (Bockelmann et al., 2009; Neela et al., 2009; Ben et al., 2019; Islam, 2019). ARGs released to the environment have been observed to persist for a long time and then eventually transfer into new hosts (Hong et al., 2018). When bacteria develop antibiotic resistance and infect humans and animals, treating these infections becomes increasingly challenging compared to infections caused by non-resistant bacteria. This difficulty in treatment is particularly evident in diseases such as pneumonia, tuberculosis, and gonorrhea, where the effectiveness of antibiotics has diminished (Islam, 2019; WHO, 2020). Consequently, this trend leads to prolonged hospitalizations, higher healthcare expenditures, economic strain on families and societies, and elevated mortality rates (Islam, 2019).

Antimicrobial resistance (AMR) among waterborne pathogens has shown a rising trend in recent years, posing a serious challenge as it renders antibiotic treatments ineffective in some cases (Lynch et al., 2013; Islam, 2019). This issue is particularly critical in developing countries, where these pathogens often cause life-threatening infections (Islam, 2019). In Bangladesh, antibiotics are widely consumed but these drugs are not disposed properly. Most of the unused residual antibiotics are discharged into wastewater drains, which might be the cause of increasing ineffectiveness of commonly used antibiotics in Bangladesh (Molla, 2019). Bangladesh faces endemic diarrheal diseases, which causes an estimated 0.1 million deaths annually (Islam et al., 2022). Enterotoxigenic *E. coli* is a leading cause of these infections in the country and the rising resistance of these bacteria to antibiotics is a major concern. Contributing factors include poor sanitation, inadequate hygiene practices, and limited access to safe drinking water (Islam, 2019). Addressing these challenges is crucial for mitigating the impact of antimicrobial resistance on public health in Bangladesh.

The problem of AMR is a major health concern, especially for developing countries (Founou et al., 2017). The World Health Organization (WHO) is actively engaged in combating this crisis, and WHO's Global Action Plan highlighting AMR surveillance as a central pillar among its core objectives (Ahmed et al., 2022). Nevertheless, there has been a notable lack of focus on AMR surveillance in water, particularly in developing nations like Bangladesh. Research on AMR in water sources primarily originates from developed regions, leaving a significant knowledge gap in the developing world (Sanganyado et al., 2019). Although a few previous studies (e.g., Neela et al., 2015; Asaduzzaman et al., 2022) have documented the presence of ARB in surface water sources in Bangladesh, research on monitoring antibiotic resistance in Bangladesh's aquatic environments remains sparse.

While our attention remains on clinical treatments and the development of new drugs, it is crucial to understand the extent and dynamics of AMR in water sources. This understanding is essential for slowing the spread of AMR and prolonging the effectiveness of life-saving antibiotics (Islam, 2019; WHO, 2020), and assess the impact of these etiological agents on human health. Research that will address

knowledge gaps in the occurrence, fate and transport, and persistence of antimicrobial resistant organisms and genes found in municipal wastewater effluent is urgently needed. The need for “a global strategy to contain resistance challenges” has been strongly proposed (Yuan et al., 2015). However, studies on antibiotic resistance have been very scarce worldwide, particularly in developing country like Bangladesh. Recently some studies have been reported the presence of antibiotics in foods such as rice, meat and dairy products in Bangladesh (Neela et al., 2012; Molla, 2019). AMR surveillance studies in Bangladesh are thus mainly focused on humans and animals. We are therefore concern about uses of antibiotic in animal husbandry and agriculture, but are unaware of developing antibiotic resistance in the aquatic environment and fishes. So far, to the best of our knowledge, no comprehensive study on antibiotic resistance in aquatic environment has been published in Bangladesh. Therefore, there is a need for research that can bridge this knowledge gap.

This study aimed to measure and characterizes the occurrences of ARB in surface water sources in and around Khulna city, Bangladesh to track hotspots of antibiotic resistance in water environments. This information could help implementing Bangladesh’s national AMR action plan 2021–2026 and could further strengthen interventions to control surface water contamination.

2. Materials and Methods

2.1 Study area

To measure ARB, water samples were obtained from five surface water sites (three from three rivers and two from two lakes) in and around Khulna city, third largest city in Bangladesh with a population of 965,000. The study sites include Daulatpur market point Bhairab river (S1); Feri Ghat point, Rupsha river (S2); Buro Darga bridge, Moyur river (S3); Bastuhara lake (S4); and Nirala lake (S5) (Fig. 1). All the sampling sites receive untreated sewage discharge through sewer drains from the populous urban and industrial areas of the city.

2.2 Sample collection, isolation and preservation of bacteria

Sampling was conducted during wet and dry seasons in 2022–2024. The sample analyses were conducted in the laboratory of Environmental Science Discipline, Khulna University. After collection, the samples were preserved in the refrigerator as soon as possible for further study. The samples were analyzed within 24hr. For preparing of culture plates, sterilized petri-dishes and medium were taken under running laminar flow as soon as possible. The medium was cooled to 40–45°C. To quantify viable bacteria, water samples were diluted with sterile phosphate buffered saline (PBS) (Neela et al., 2015). A ten-fold serial dilution was made. CFU/ml of isolates were counted according to Reynolds et al. (2005). The plates were then incubated at 37.5°C for 24 hours.

Isolation of the discrete bacterial colonies was carried out immediately after counting of the colonies. On the basis of their morphological characteristics, several different colonies were selected for this purpose. The chosen colonies were identified and their characteristics were analyzed based on various criteria, namely form, elevation, shape, size, texture, colony surface marking, margin type, color, translucency or appearance etc.

The isolates were then preserved in glycerol stock. At first 700 µl of 33% glycerol was taken in each 1.5ml eppendorf tubes and sterilized by autoclaving. The overnight incubated broth culture of the isolates of about 700 µl was added in each eppendorf tubes separately. The cap of the eppendorf tubes were covered by parafilm and stored at -20°C.

2.3 Antibiotic susceptibility test

For ARB analysis, at first antimicrobial susceptibility tests were performed. The antibiotics used in this study were Ampicillin (AMP), Ciprofloxacin (CIP), and Ceftriaxone (CTX). Antibiotic resistant bacteria were cultured in nutrient broth at 25°C for 24 hrs with shaking. Antibiotic sensitivity test was done by Kirby-Bauer disc diffusion method (Adomako et al., 2021), where antibiotic discs (Bioanalyse, Turkey) were placed on an agar plate on bacterial lawn with cell suspension with a density of MacFarland No. 0.5 was spread on agar medium. Plates were incubated overnight at an incubation temperature of 37°C. The antibiotic discs used were Ampicillin 10µg, Ciprofloxacin 5µg and Ceftriaxone 30µg. The diameters of inhibition zones were measured in millimeters and recorded for each of the antibiotics (Adomako et al., 2021). The size of these zones depends on the antibiotic's effectiveness in halting the growth of bacterial colonies.

2.4 Identification of resistant isolates

The presence of the ARG in water samples were analyzed by PCR in the laboratory of Invent Technologies Ltd, Banani, Dhaka. For PCR analysis at first DNA extraction was done. Cells were harvested by centrifugation at 3000 × g for 10 min and then DNA was extracted (Rahman et al. 2008). Then DNA sequencing was done to identify resistant bacteria by 16S rRNA gene analysis. The 16S rRNA gene sequence was about 1,550 base pair (bp) long. PCR amplification of 16S rRNA gene was conducted using set of primers 27f (5'-AGAGTTTGATCCTGGCTCAG-3') and 1492r (5'-CTACGGCTACCTTGTACGA-3') as performed by Neela et al. (2015), which is the most widely used primer for species-level identification (Frank et al., 2008). DNA extraction was performed using automated DNA extractor, Maxwell 16, Promega, USA. Online similarity searching was performed using the Basic Local Alignment Search Tool (BLAST) available at the National Center for Biotechnology Information website (NCBI, <http://www.ncbi.nlm.nih.gov/>); BLAST search was carried out on using the NCBI nucleotide database "16S rRNA sequences (Bacteria and Archaea)" with the program selection optimized for "Highly similar sequences (MegaBLAST)".

3. Results

3.1 Spatiotemporal variability of ARB

The viable bacterial number ranged from $1.6 \pm 0.07 \times 10^4$ to $2.4 \pm 0.14 \times 10^5$ cfu/ml in the surface waters in and around Khulna city (Table 1). The bacterial counts in the lakes were one order of magnitude higher than those in the rivers. Overall, of the 160 strains from samples of the surface water sources (32 from each of the sampling sites), 9.4–18.8%, 75.0–87.5% and 65.6–78.1% of them showed resistance to AMP, CIP and CTX, respectively (Table 1). AMP showed a comparatively lower resistance rate than that of CIP and CTX. No distinct pattern of variability was observed among the sampling sites. The prevalence of ARB was observed to be slightly higher during the wet seasons (June–September) compared to the dry seasons (Nov–May)(Fig. 2).

Table 1
Viable count of antibiotics resistant bacteria in surface water, Khulna city

Sampling site	Total viable count cfu/ml	AMP resistant (%)	CIP resistant (%)	CTX resistant (%)
Bhairab river (S1)	$1.6 \pm 0.07 \times 10^4$	15.6	75	71.9
Rupsha river (S2)	$4.2 \pm 0.09 \times 10^4$	9.4	78.1	65.6
Moyur river (S3)	$7.8 \pm 0.11 \times 10^4$	12.5	81.2	68.8
Bastuhara lake (S4)	$2.4 \pm 0.14 \times 10^5$	18.8	87.5	78.1
Nirala lake (S5)	$1.3 \pm 0.08 \times 10^5$	12.5	84.4	71.9

3.2 Identification of ARB

To identify species of the resistant bacteria, we selected 56 strains randomly that were resistant to the antibiotics AMP, CIP, and CTX. The PCR amplification of 16S rRNA gene analysis results showed that antibiotic resistant *Shigella flexneri* was the most dominant in surface water 17.9% (10/56) followed by *Escherichia fergusonii* 12.5% (7/56), *Proteus mirabilis* 10.7% (6/56) and *Enterobacter quasiormaechei* 8.9% (5/56) (Table 2). Different other species of *Enterobacter*, such as *Enterobacter mori* 7.1% (4/56), *Enterobacter sichuanensis* 5.4% (3/56) were also showed substantial dominance in the surface water sources.

The taxonomic composition of the antibiotic resistant bacteria community at the genus level showed that *Enterobacter* (23.5%), *Shigella* (20.6%), and *Escherichia* spp (14.7%) were found the most abundant both in the rivers and lakes water (Fig. 3).

Table 2
Identification of antibiotic resistant bacteria by 16S rRNA gene sequencing

Sampling month	Isolate ID	Length of sequences (bp)	Identified putative species having closest relation	Similarity (%)	Gene bank accession number
Feb, 2022	08022022D1	1255	<i>Enterobacter cloacae</i>	96.9	NR_113615.1
Feb, 2022	08022022B1	1271	<i>Klebsiella pneumoniae</i>	98.2	NR_117683.1
Feb, 2022	08022022B2	1130	<i>Escherichia fergusonii</i>	93.7	NR_027549.1
Feb, 2022	08022022B2	1274	<i>Bacillus aerius</i>	98.3	NR_118439.1
Jun, 2022	03062022M1	1111	<i>Shigella flexneri</i>	95.9	NR_026331.1
Jun, 2022	03062022M2	1197	<i>Shigella flexneri</i>	97.7	NR_026331.1
Jun, 2022	03062022M3	1183	<i>Escherichia fergusonii</i>	96.6	NR_027549.1
Jun, 2022	03062022N1	888	<i>Shigella flexneri</i>	98.9	NR_026331.1
Jun, 2022	03062022N2	1151	<i>Escherichia fergusonii</i>	95.8	NR_027549.1
Jun, 2022	03062022M4	1191	<i>Escherichia fergusonii</i>	96.8	NR_027549.1
Aug, 2022	200820225M1	438	<i>Shigella dysenteriae</i>	99.1	NR_026332.1
Aug, 2022	200820222M2	675	<i>Escherichia fergusonii</i>	98.4	NR_027549.1
Aug, 2022	20082022R1	300	<i>Enterobacter quasiormaechei</i>	96.6	NR_180451.1
Aug, 2022	20082022D1	835	<i>Planococcus citreus</i>	92.0	NR_113814.1
Sep, 2022	12122022B2	1226	<i>Enterobacter quasiroggenkampii</i>	86.5	NR_179166.1
Sep, 2022	12122022B3	1280	<i>Enterobacter quasiroggenkampii</i>	82.8	NR_179166.1
Sep, 2022	12122022D1	1359	<i>Klebsiella pneumoniae</i>	96.4	NR_037084.1
Dec, 2022	12122022B1	1193	<i>Enterobacter cloacae</i>	96.9	NR_028912.1
Dec, 2022	12122022N1	1297	<i>Enterobacter sichuanensis</i>	85.1	NR_179946.1
Dec, 2022	12122022M1	1197	<i>Klebsiella pneumoniae</i>	97.8	NR_117683.1
Dec, 2022	12122022R2	1280	<i>Enterobacter sichuanensis</i>	87.7	NR_179946.1
Mar, 2023	02032023M2	1176	<i>Citrobacter freundii</i>	94.2	NR_114345.1

Sampling month	Isolate ID	Length of sequences (bp)	Identified putative species having closest relation	Similarity (%)	Gene bank accession number
Mar, 2023	02032023M2	1157	<i>Klebsiella variicola</i>	92.3	NR_025635.1
Mar, 2023	02032023M4	1217	<i>Exiguobacterium indicum</i>	92.8	NR_042347.1
May, 2023	19052023N1	1242	<i>Klebsiella pneumoniae</i>	96.9	NR_119276.1
May, 2023	19052023N2	1236	<i>Shigella flexneri</i>	97.8	NR_026331.1
May, 2023	19052023N3	1026	<i>Metabacillus indicus</i>	85.8	NR_029022.1
May, 2023	19052023N4	1121	<i>Staphylococcus caprae</i>	91.4	NR_119252.1
May, 2023	19052023D1	1316	<i>Escherichia fergusonii</i>	96.7	NR_027549.1
May, 2023	19052023D2	1258	<i>Shigella flexneri</i>	97.4	NR_026331.1
May, 2023	19052023B1	1250	<i>Escherichia fergusonii</i>	97.5	NR_074902.1
May, 2023	19052023B2	1274	<i>Shigella flexneri</i>	96.3	NR_026331.1
May, 2023	19052023B3	1138	<i>Bacillus pseudomycooides</i>	78.8	NR_113991.1
May, 2023	19052023R1	1228	<i>Enterobacter sichuanensis</i>	97.9	NR_179946.1
May, 2023	19052023R2	1137	<i>Shigella flexneri</i>	95.0	NR_026331.1
May, 2023	19052023R3	1337	<i>Shewanella xiamenensis</i>	96.9	NR_116732.1
May, 2023	19052023R4	1176	<i>Shewanella xiamenensis</i>	90.9	NR_116732.1
Nov, 2023	03112023B1	1227	<i>Enterobacter mori</i>	95.5	NR_146667.2
Nov, 2023	03112023D1	429	<i>Enterobacter quasiroggenkampii</i>	94.1	NR_179166.1
Nov, 2023	03112023M1	1227	<i>Staphylococcus hominis</i>	97.8	NR_041323.1
Nov, 2023	03112023N1	1269	<i>Enterobacter mori</i>	97.6	NR_146667.2

Sampling month	Isolate ID	Length of sequences (bp)	Identified putative species having closest relation	Similarity (%)	Gene bank accession number
Nov, 2023	03112023N2	765	<i>Enterobacter quasiroggenkampii</i>	94.5	NR_179166.1
Nov, 2023	03112023N3	1179	<i>Enterobacter mori</i>	95.0	NR_146667.2
Nov, 2023	03112023R1	1149	<i>Exiguobacterium arabatum</i>	96.4	NR_178375.1
Nov, 2023	03112023R3	1216	<i>Exiguobacterium qingdaonense</i>	97.3	NR_181609.1
Nov, 2023	03112023D3	1189	<i>Enterobacter mori</i>	93.7	NR_146667.2
Feb, 2024	18022024B1	1069	<i>Shigella flexneri</i>	87.6	NR_026331.1
Feb, 2024	18022024B3	1275	<i>Shigella flexneri</i>	88.3	NR_026331.1
Feb, 2024	18022024D1	1239	<i>Proteus mirabilis</i>	97.6	NR_113344.1
Feb, 2024	18022024D4	1254	<i>Proteus mirabilis</i>	99.0	NR_113344.1
Feb, 2024	18022024M1	1179	<i>Proteus mirabilis</i>	97.4	NR_113344.1
Feb, 2024	18022024M3	1217	<i>Proteus mirabilis</i>	98.7	NR_113344.1
Feb, 2024	18022024N2	964	<i>Shigella sonnei</i>	86.5	NR_104826.1
Feb, 2024	18022024N4	1234	<i>Shigella flexneri</i>	87.8	NR_026331.1
Feb, 2024	18022024R2	1254	<i>Proteus mirabilis</i>	98.4	NR_113344.1
Feb, 2024	18022024R3	1208	<i>Proteus mirabilis</i>	97.5	NR_113344.1

4. Discussion

In this study, we investigated the presence of ARB in three rivers and two lakes to assess the presence of antibiotic resistance in surface water sources in Khulna city, Bangladesh. The rivers and lakes were selected to cover two different freshwater ecosystems. The obtained viable bacterial number ranged from $1.6 \pm 0.07 \times 10^4$ to $2.4 \pm 0.14 \times 10^5$ cfu/ml, which is in agreement with the study of Neela et al. (2015) based on aquaculture ponds in Bangladesh. The bacterial counts in the lakes were one order of magnitude higher than those in the studied rivers, which is because the rivers have tidal influence and more dilution effect.

The study reveals that 9.4–18.8%, 75.0–87.5% and 65.6–78.1% of the isolate showed resistance to AMP, CIP and CTX, respectively, which are comparable to other studies, for instance, 17.95% and 12.3% resistance to AMP respectively by Neela et al. (2012) and Navarro et al. (2023); 54.1% resistant to CTX (Ahmed et al., 2022); and 55.5% resistant to CIP (Neyestani et al., 2023). In our study, CIP and CTX

showed a comparatively higher resistance rate than that of AMP. This high resistance rate can be attributed to the high consumption of ciprofloxacin and Ceftriaxone by Bangladeshi people. The widespread prevalence of ARB across the surface water sources might be because of continuous discharge of antibiotic residues from nearby hospitals, unregulated farm practices and other anthropogenic activities.

We identified an elevated prevalence of ARB during the wet periods compared to the dry periods, aligning with the research of Di Cesare et al. (2017). Their study revealed increased abundances of antibiotic resistance genes during rainy seasons in a large subalpine river in Italy. The increase in the abundance of ARB during rainy season indicated that rainwater might act as a reservoir that can facilitate the transmission of ARB in the aquatic environment.

The taxonomic composition of the ARB showed that antibiotic resistant *Enterobacter* spp (23.5%), *Shigella* spp (20.6%), and *Escherichia* spp (14.7%) were most abundant both in the rivers and lakes water. This is in agreement with the previous study (Neela et al., 2015) that reported *Enterobacter* spp (16.7%), *Shigella* spp (5.6%) are the two dominant ARB, among others. Zahara et al. (2022) found *Shigella* spp in rivers in Banda Aceh, Indonesia; Crettels et al. (2023) found *Escherichia* spp in freshwater in Belgium; and Bartholin et al. (2023) identified *Enterobacter* spp in the Environmental Water Sources from Southern Chile as one of the dominant ARB.

Enterobacter, a genus within the Enterobacteriaceae family, is primarily associated with infections related to healthcare. These resilient bacteria are known culprits in community-acquired ailments such as urinary tract infections, respiratory infections, osteomyelitis, and endocarditis. Commonly isolated from respiratory sputum, surgical wounds, and blood samples, *Enterobacter* species pose a significant challenge in intensive care units (Ramirez & Giron, 2023). Alarming, *Enterobacter* strains are evolving resistance to many antibiotics that were previously effective, exacerbating treatment difficulties. Recognizing this threat, the WHO in 2017 identified carbapenem-resistant Enterobacteriaceae (CRE) as a critical priority for urgent antibiotic development (Ramirez & Giron, 2023). Adding to the concern, current study also supports that *Enterobacter* spp is the most prevalent ARB in surface waters in Khulna city. This finding underscores the urgent need for comprehensive strategies to combat the spread of *Enterobacter* infections and the development of novel antibiotics to address their increasing resistance.

The study reveals that ARB are highly prevalent in surface water samples. This prevalence could result in increased human exposures to antimicrobial resistance (AMR). A surveillance strategy to monitor the prevalence of AMR and implementing effective intervention measures are needed to reduce the spread of AMR to humans from the environment. It is crucial to investigate and quantify human exposure to AMR through water and assess its health impacts (Islam et al., 2021). Quantitative microbial risk assessment (QMRA) has emerged as a popular probabilistic modeling technique used to estimate health risks linked to exposure to waterborne pathogens (Abel and Taylor, 2018; Dias et al., 2019; Islam, 2019). QMRA seeks to evaluate the risk of adverse health effects resulting from human exposures to infectious microbes (Islam & Islam, 2020). Since antibiotics are extensively used now a day, risk assessments of

human exposure to AMR in the environment have been attracted increasing attention (Ashbolt et al., 2013; Ben et al., 2019). Development of a framework to assess the risks associated with ARB and ARGs during reuse is urgently needed. However, worldwide so far no comprehensive QMRA has been conducted for ARB. Ashbolt et al. (2013) first stated the opportunity to use QMRA for AMR. Afterwards, Schijven et al. (2015) tried to provide a first example of such a QMRA for ARB, as also discussed by Wuijts et al (2017). However, such a QMRA is not easy. While for ordinary bacteria, acquiring the dose response data is relatively easier, for ARB/ARG this is not the case. Consuming ARB is not necessarily a problem. It becomes a problem when other bacteria make a person sick and the antibiotics don't work anymore because resistance can develop in the environment amidst low concentrations of antibiotics and the genes can also be transmitted from one bacterium to another (Rose et al., 2023). It is very difficult to account for this in the QMRA of ARB. So developing a QMRA for AMR, which is difficult but may not impossible, can be a next step, which was beyond the scope of this study.

5. Conclusion

In recent years, concern has increased about the occurrence of ARB and ARGs that could make it more difficult to treat certain infections in animals and people. This study provides detail information on the dynamics of ARB in surface water. The monitoring results revealed that the surface waters in and around Khulna city could be considered as a reservoir for the ARB, emphasizing the importance for routine monitoring and surveillance activities. These results underscore the pivotal role surface water can assume in propagating AMR. Urgent action is required to raise awareness among stakeholders and communities about the detrimental effects of unregulated wastewater disposal on the environment. The findings of this study will help determine the ARB reduction target values in wastewater treatment. Research results should improve our understanding of the nature, extent, selection, and removal of ARB and ARGs in municipal wastewater effluent. The study provides valuable information for the management of surface water and to ensure safe uses of the rivers and lakes. The findings will also help implementing Bangladesh's national AMR action plan 2021–2026 and contributing to reach the Sustainable Development Goal (SDG) 6 to improve water and sanitation.

Declarations

All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors.

Data availability

Most of the data generated are included in this article. Additional data will be made available on reasonable request.

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by M.M. Majedul Islam, Md Atikul Islam and Abul Farah Md. Hasanuzzaman. The first draft of the manuscript was written by M.M. Majedul Islam and all authors commented on previous versions of the manuscript. All authors have read and agreed to the published version of the manuscript.

Ethical Approval

Not applicable (this study did not recruit any human/animal subjects).

Consent to Participate

Not applicable (this study did not recruit any human subjects).

Consent to Publish

Not applicable (this study is not published previously).

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Figures

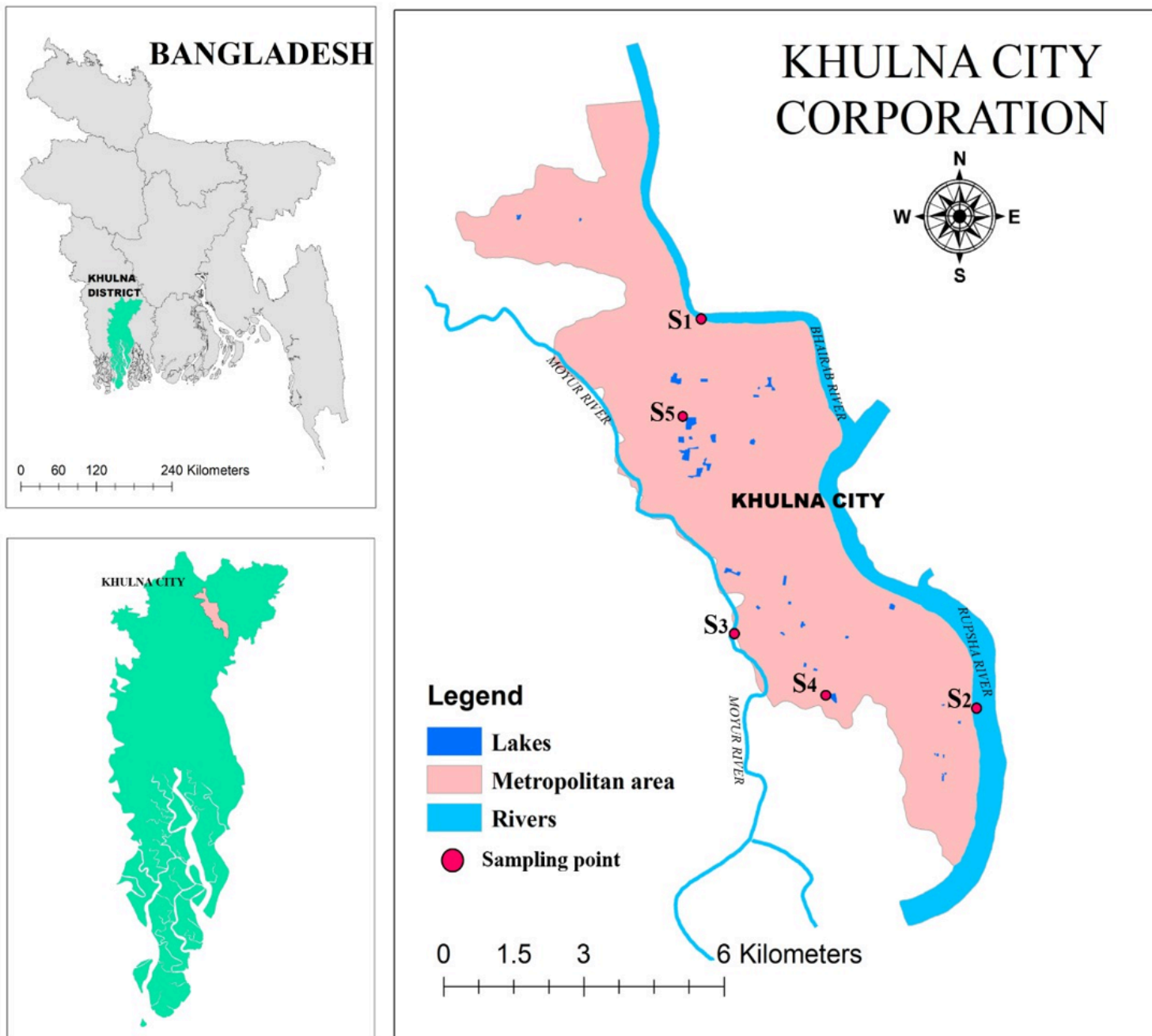


Figure 1

Sampling sites in the Rupsha, Bhairab and Moyur rivers; and Bastuhara and Nirala lakes in Khulna city.

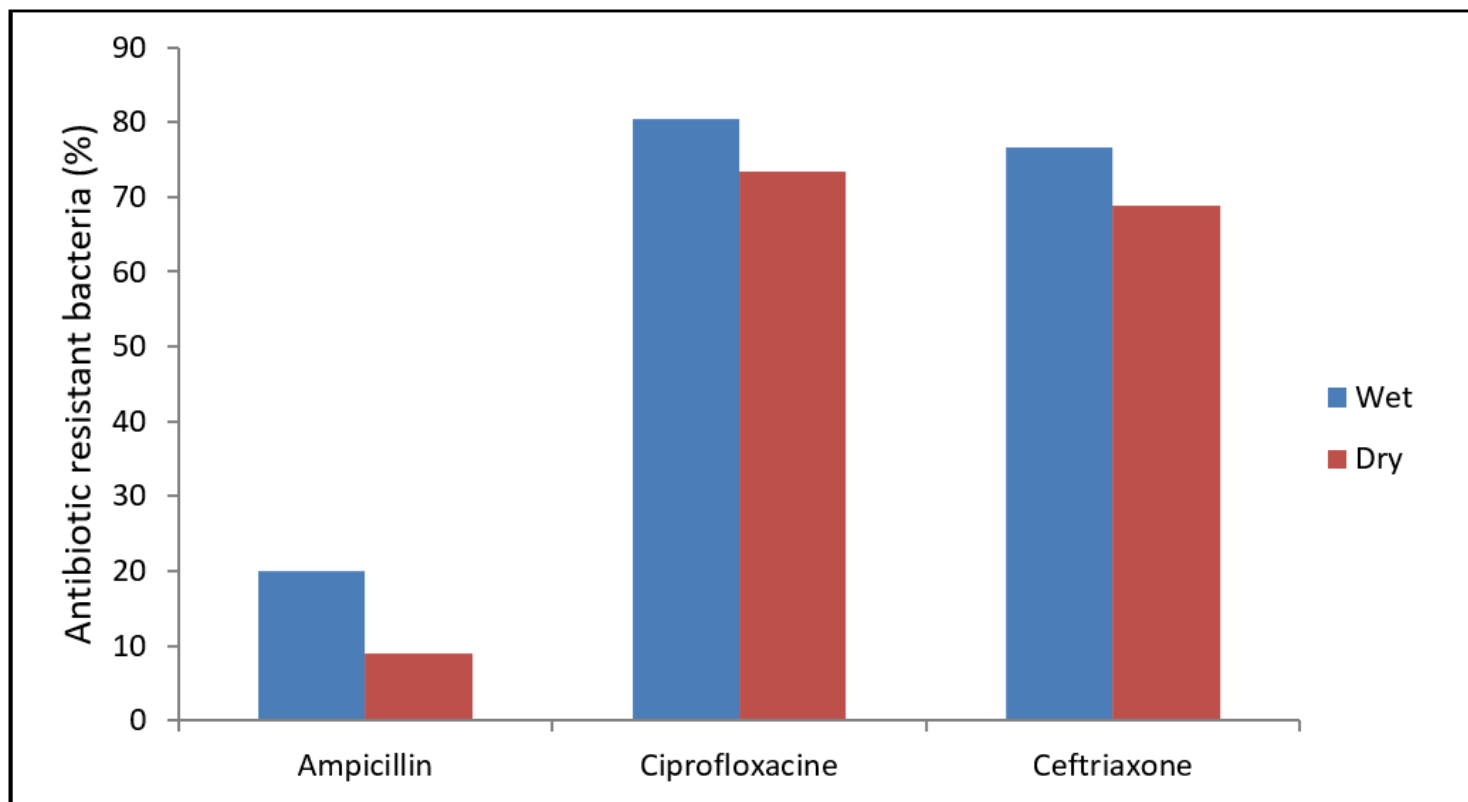


Figure 2

Seasonal variability of antibiotic resistant bacteria in the rivers and lakes in Khulna city

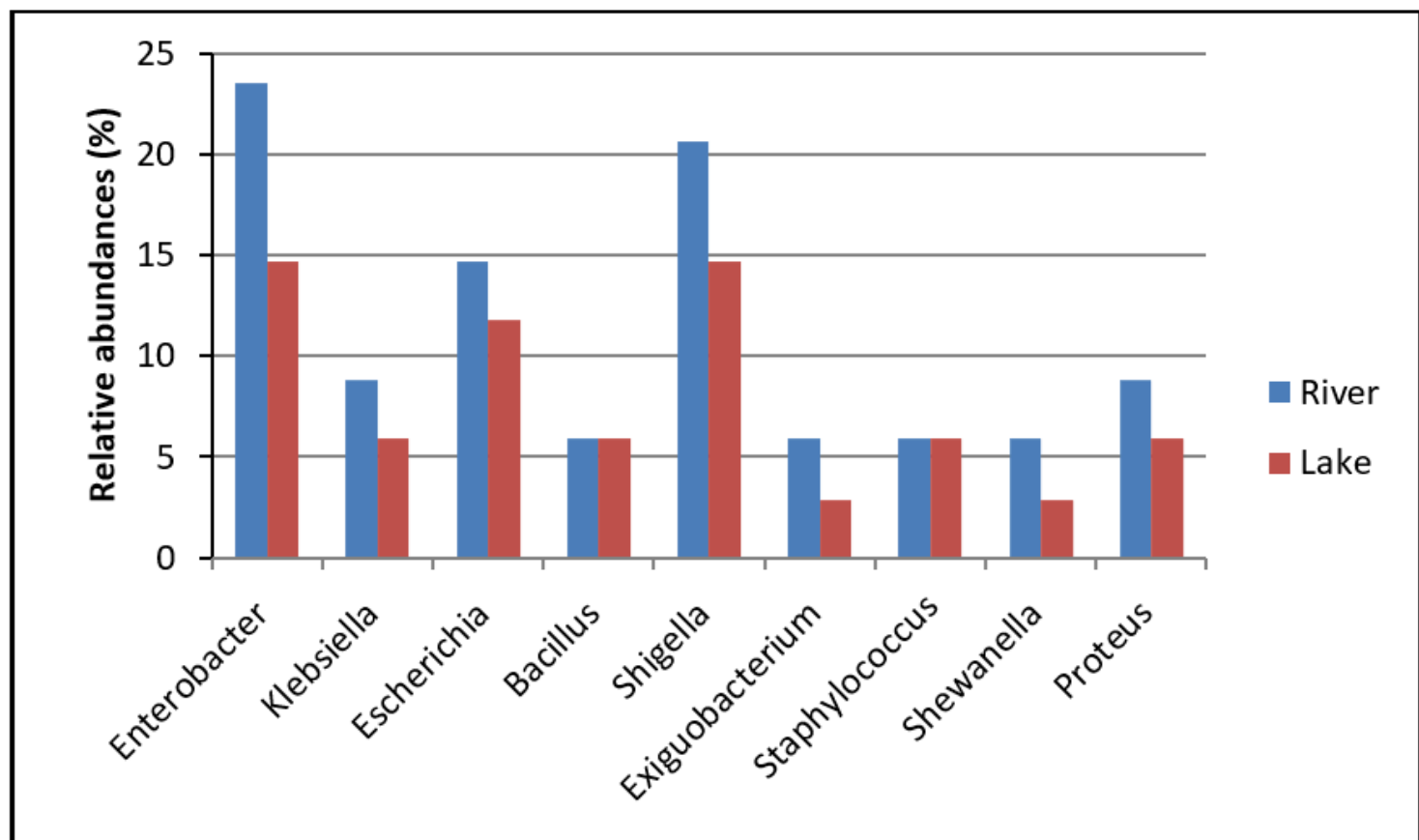


Figure 3

Taxonomic composition and relative abundance (%) of the bacterial community at the genus level for the rivers and lakes based on the PCR amplification of 16S rRNA gene analysis results. Only dominant genera with more than one time occurrences are shown.